

```

(0) - genotyping_in_silico: 5 windows
(1) -> + 0: bash-
(2) -> + 1: tree_calc*
(3) -> + 2: bash
(4) -> + 3: bash
(5) -> + 4: bash
(6) -> + 1: 1 windows
(7) -> + 0: bash*
(8) -> + 2: 2 windows (attached)
(9) -> + 0: http-
(M-a) -> + 1: [tmux]*Z

1 (sort: index)
  x $ t 2 3 Z h < 1
  p l R q I * % R . *
  B 7 ; z C I + , ( 7 H u
  - 2 3 w T A G J f
  - U R S z 9 # [ 0 Z
  f V b j + - z U 9 D x
  d z B g ; A D P ^ 3 0
  . : z 8 W u 7 H ^ Q Y
  : U w N y v @ M K Y h
  Q @ j r b R A x > j ) W 8 1
  l # 0 B 1 X 7 0 < \
  x I 2 G 0 [ 0 1 > 0 y + h
  < m : 2 T 1 D ) u Y $ a
  - + [ N z + y b K > 7 U # * $
  q A ' Y ( N < H b > / < E : F
  C r S I b @ 4 G - 1 D , 6
  X j ! C 0 f D 2 7 - = q ;
  - a 2 % u T : 0 1 / 5 b > /
  u Z M k 0 P k d m 8 d \ n
  X * : j ( Z B p # 7 W N b X
  T h L ( ) 5 d K / j h s m C

0[ 0.0%] 4[ 0.0%] 8[ 3.3%] 12[ 0.0%] 14[
1[ 0.0%] 5[ 0.0%] 9[ 0.0%] 13[ 0.0%] 15[
2[ 0.0%] 6[ 2.0%] 10[ 0.0%] 16[
3[ 2.6%] 7[ 0.0%] 11[ 0.0%] 17[
Mem[ ] 842M/236G Task
Swp[ ] 14.8M/1024M Load
Upt[ ]

Main 1320
PID USER PRI VIRT RES SHR S C
101546 ubuntu 20 1 5852 3792 3796 S
294595 ubuntu 20 6644 2712 2256 S
294334 ubuntu 20 0 9072 5028 3568 S
F1Help F2Setup F3Search F4Filter F5Tree F6Sort F7

[2] 0:htop- 1:[tmux]*Z

```

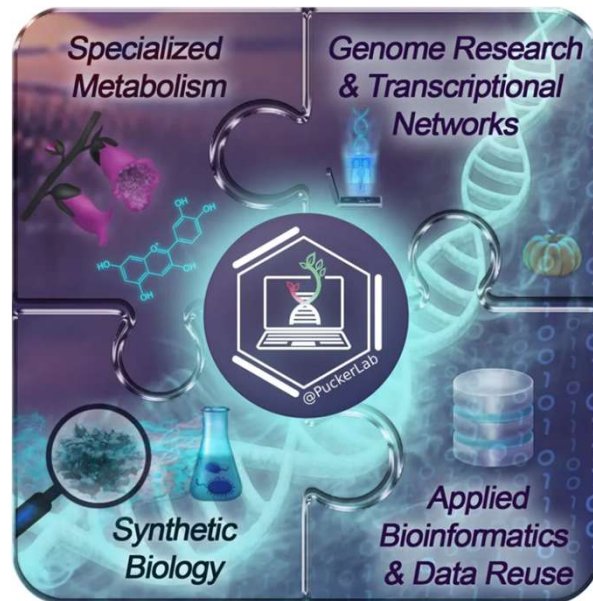
Samuel N. Meckoni & Nancy Choudhary

Command-line interface (CLI) basics for life scientists

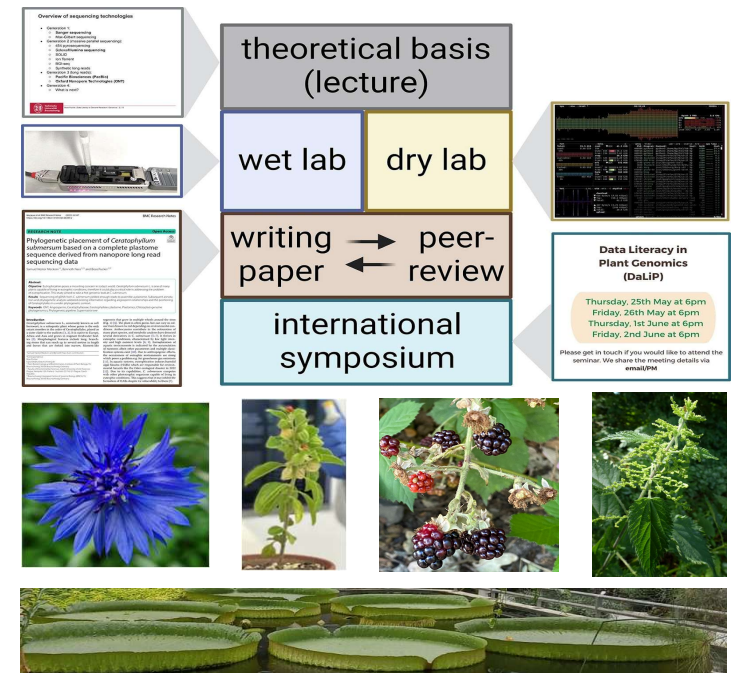
Plant Biotechnology & Bioinformatics (PBB)

Group overview

- Flavonoids:
Anthocyanins,
Flavonols, Flavones,
Proanthocyanidins
- Carotenoids
- Withanolides
- ...



Tools: KIPES, NAVIP, MGSE,
MYB annotator, DuplyliCate...



Wolff *et al.*, 2023: 10.1515/jib-2023-0033 2

Dassow *et al.*, 2025: 10.1101/2025.01.31.635922



Course Overview

CLI basics for life scientists

- Basics of the Unix command line interface
- Unix file system
- Manipulating files and directories
- Combining bash commands
- Using these commands for biological insights



Samuel



Nancy



Boas

Availability of course material

- All materials will be shared at the end of this course.

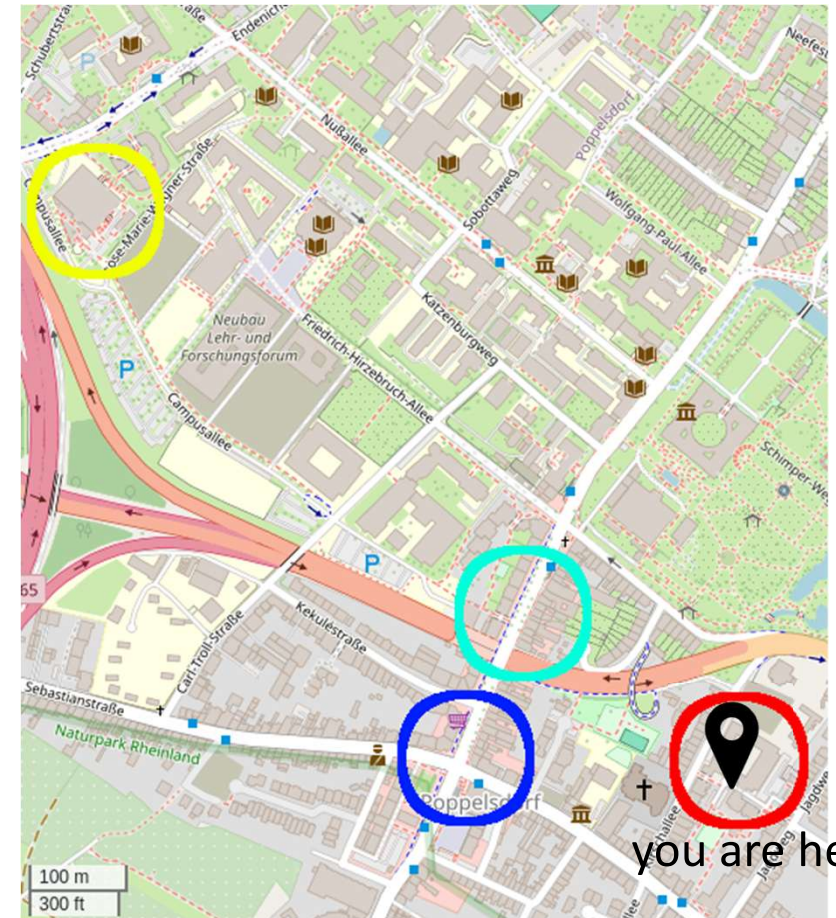
https://github.com/PuckerLab/teaching/tree/main/deNBI2026_CLIbasicsforlifescientists

Organisational I

- 10 AM -5 PM, 1 hour lunch break
- Tuesday: lunch break from 1 PM - 3 PM
 - PhD defense at 2 PM in 2.007
- Friday until 1 PM

Organisational II

- Mensa (15 mins walk)
- Local food stores
- Supermarkets



Recap

- Using CLI commands for gaining biological insights
- How to prepare data for transcriptomics analysis

Long runs

- Some runs will take a long time to finish (hours < days < weeks)
- ssh connections usually don't last as long → the run will be interrupted
- Solution: background process

Background and foreground

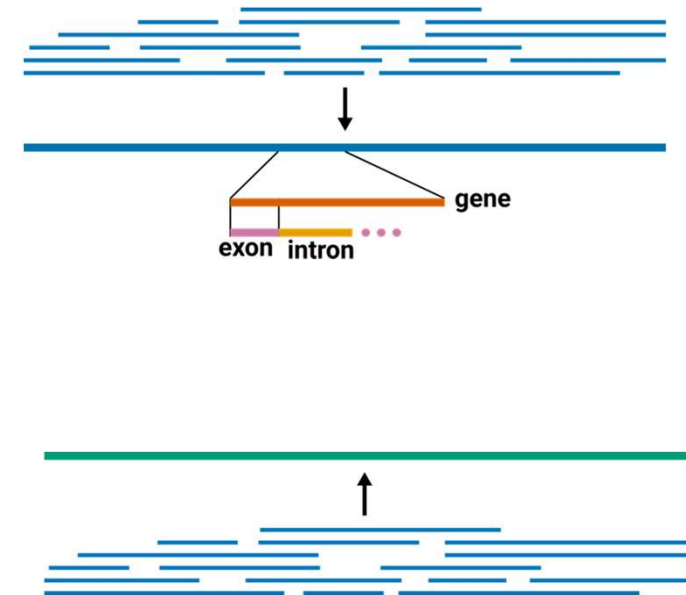
- Ctrl (/Strg) + Z to stop the command
- fg = bring job to foreground
- bg = bring job to background
- or append a "&" at the end of the command before running it

Alternative : tmux

- Multiplexer
- Multiple CLI windows and panes in one session
- Keeps running in the background, even if you stop the ssh connection

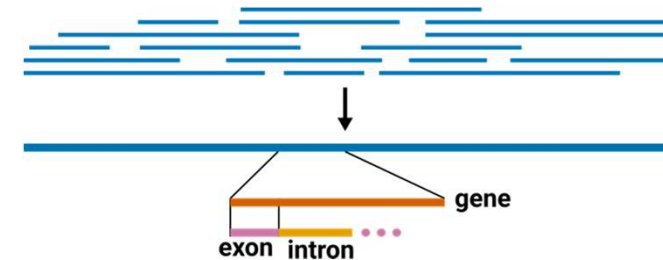
A typical genomics workflow

- Assembly
- Annotation
- Read mapping



A typical genomics workflow

- Assembly
- Annotation
- Read mapping

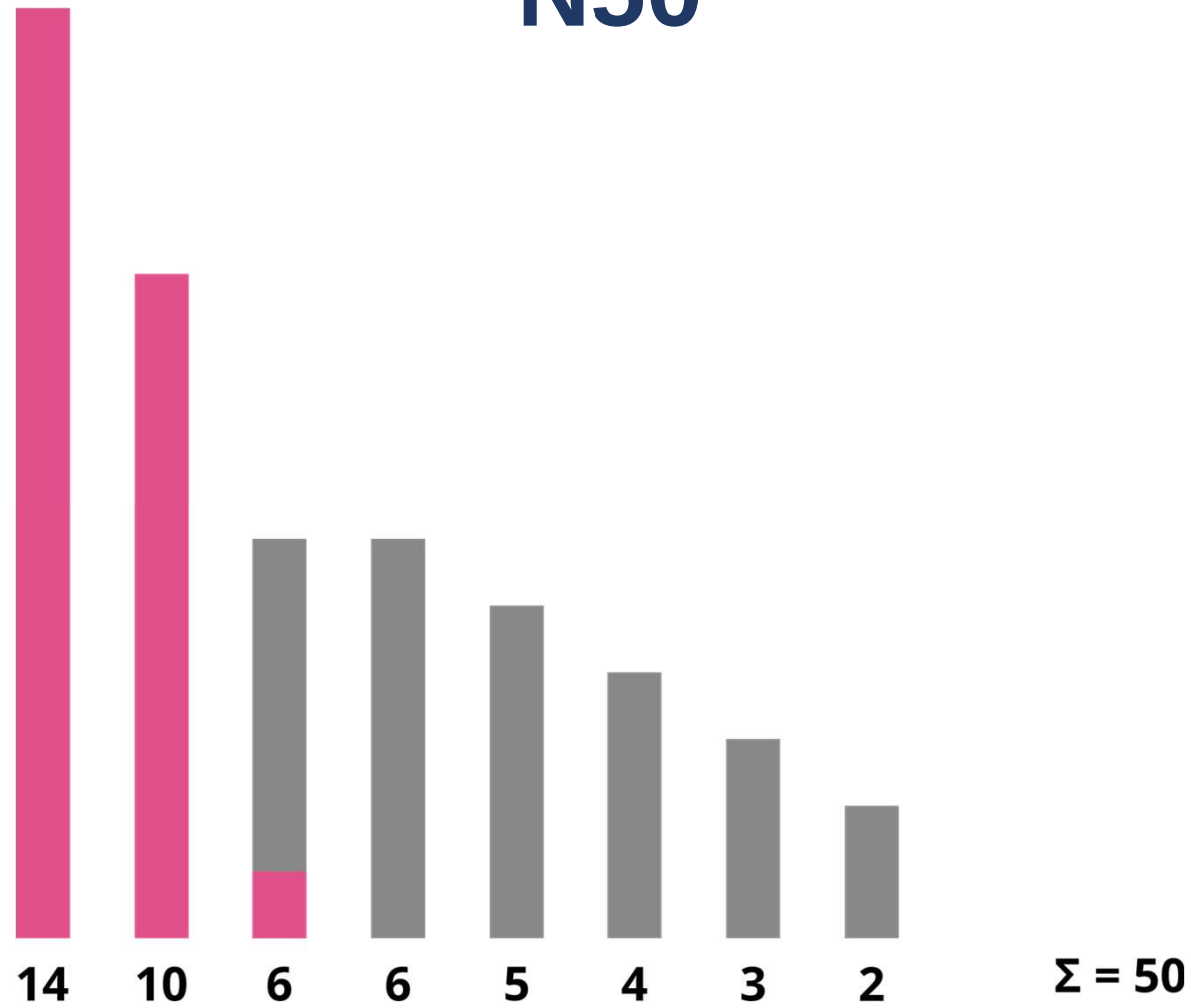


Metrics

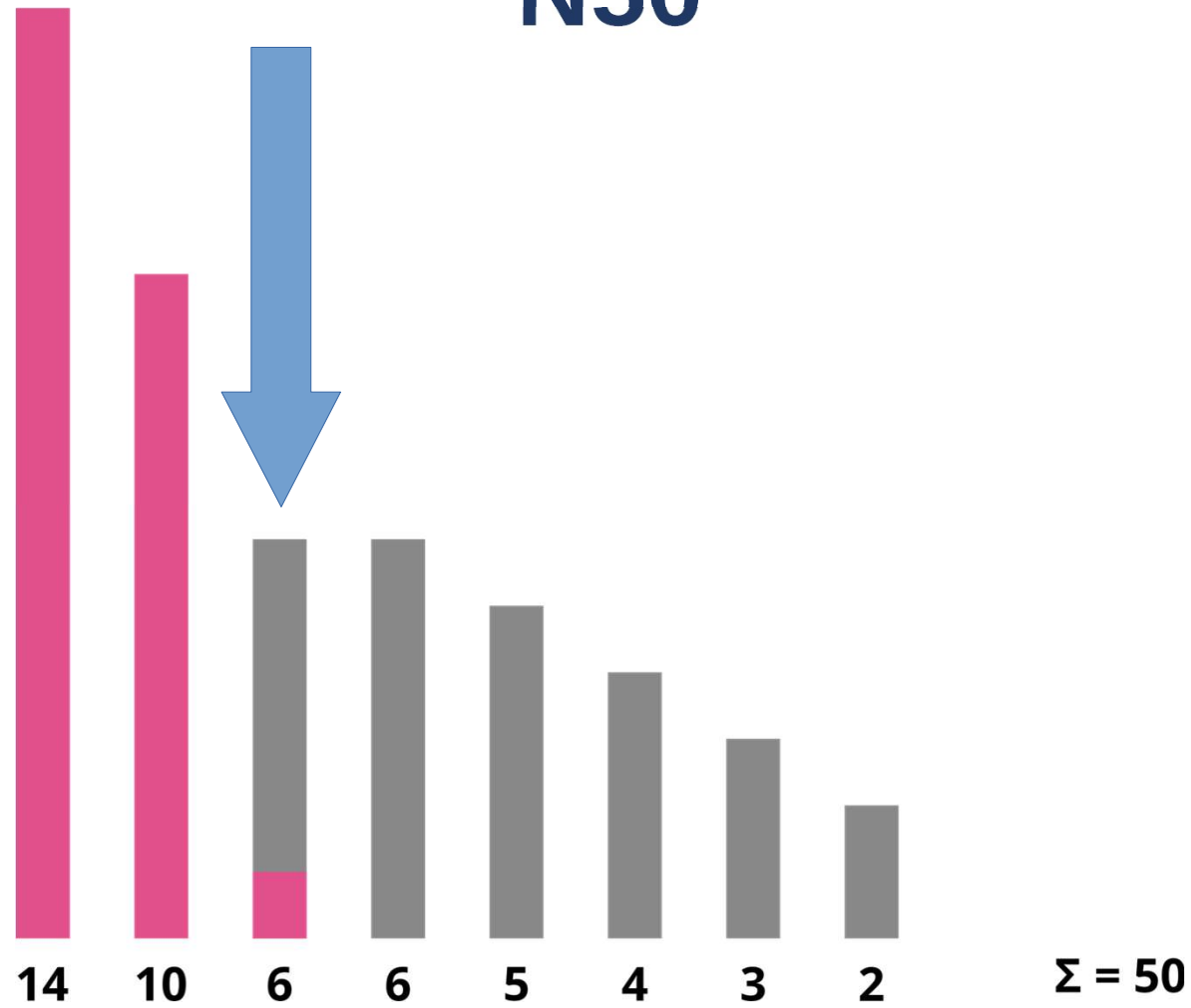
- Amount of reads
- Coverage depth



N50



N50



Exercises

- Task 3.1: Download additional data from the shared VM:
 - Genome contig:
`/vol/data/task3/daUtrGibb.v01.asm.ptg000007l.fasta`
 - Reads subset:
`/vol/data/task3/subsample_u_gibba.0.01.2.fastq`

Exercises

- Task 3.2: install seqkit with apt and get some statistics from a fastq and a fasta file from the previous day:
 - No. of sequences
 - Longest sequence
 - How many sequences are longer than 1 Mbp
 - Can you get the N50?

Exercises

- Task 3.3: Use minimap2 to map subsample_u_gibba.0.01.fastq against daUtrGibb.v01.asm.ptg000001l.fasta
- Task 3.4: Use samtools to sort and index file resulting in an alignment file

Exercises

- Task 3.3: Use minimap2 to map subsample_u_gibba.0.01.fastq against daUtrGibb.v01.asm.ptg000001l.fasta
- Task 3.4: Use samtools to sort and index file resulting in an alignment file
- Task 3.5: Redo task 3.3 and 3.4, but this time in one command

Exercises

- Task 3.6: Calculate the read coverage with samtools
 - Which position has the highest and which has the lowest coverage?
 - What is the average coverage (arithmetic mean and median)?

Scripts

How to execute multiple commands?

- scripts
- write multiple commands into a file
- “execute” the file once
- execute like this
 - `bash /path/to/script.sh`
 - `/path/to/script.sh`
 - `source /path/to/script.sh`
 - ...



Loops

- **while loop:**
while read line; do
 echo \$line
done < ./ids
- **for loop:**
for i in {1..3}; do
 echo "Hi"
done

First small script

- Task 4.1: write a script that creates a new directory. In this directory, it should create a file that contains 10 times the output of the “fortune” command. Then it should print out this file.

Variables

- variables ...
- define a variable
var1="text"
var2=0
- program output as variable:
var3=\$(fortune)
- call a variable with "\$" and the name of the variable
e.g.: echo \$var2

→ Also works in the CLI



Positional arguments/variables

- Can be called in the script with \$1 or \$2, \$3, ...
- Equals the positional input arguments
- Example:

```
sam@debian-nds: ~  
(base) sam@debian-nds:~$ cat echo_script.sh  
echo $2 was given after $1  
(base) sam@debian-nds:~$ bash echo_script.sh 2 hi  
hi was given after 2  
(base) sam@debian-nds:~$
```